

# Audio-Visual Learning

*Free supervision from synchronised video — contrastive SSL, AVHuBERT, AudioCLIP, and temporal alignment*

**9 min**

Duration

**B3 – Apply**

Bloom's

**L3 – Intermediate**

Level

**TH-000007 & TH-000010**

Prerequisite

# Session Overview

*Natural synchrony as a training signal — the most abundant free supervisory signal in existence*

## What You Will Learn

- Why audio and video are naturally paired learning signals
- The McGurk Effect as evidence for human audio-visual fusion
- Self-supervised audio-visual learning methods (AVC, AVS, MAVM)
- Contrastive learning with InfoNCE / NTXent objective
- AVHuBERT: masked audio-visual pre-training
- AudioCLIP: three-way audio-image-text contrastive learning
- Sound source localisation: seeing what you hear
- Temporal alignment challenges and engineering solutions

## Concept at a Glance

- Duration: 9 minutes
- Bloom's Level: B3 – Apply
- Difficulty: L3 – Intermediate
- Module: 4.3 Multimodal Fusion and Training
- Prerequisites: Audio Features (TH-000007), Fusion Strategies (TH-000010)
- Next: Training Multimodal Systems (TH-000012)

# Learning Objectives

*By the end of this session you will be able to:*

**9 min**

Duration

**B3 – Apply**

Bloom's

**L3 – Intermediate**

Level

**TH-000007 & 000010**

Prerequisite

- 1 Explain why audio and video are naturally paired and what 'free supervision' means
- 2 Describe self-supervised audio-visual pre-training pretext tasks (AVC, AVS, MAVM)
- 3 Apply the InfoNCE contrastive loss to the audio-visual correspondence problem
- 4 Compare AVHuBERT and AudioCLIP architectures and their key capabilities
- 5 Identify downstream tasks enabled by audio-visual pre-training
- 6 Recognise temporal misalignment challenges and the engineering strategies to mitigate them

## SECTION 1 — THE NATURAL PAIRING

# Why Audio and Vision Belong Together

*The physical world generates audio and visual signals simultaneously — this is free supervision*

# The Natural Co-Occurrence of Audio and Vision

*A dog barking, a guitar playing, a car accelerating — the visual event causes the audio event*

## The Physics of Free Supervision

The physical world generates audio and visual signals simultaneously from the same source. This causal relationship — visual event produces audio event — provides a virtually unlimited training signal. Every video ever recorded is a labelled audio-visual pair. No human annotation needed.

## The McGurk Effect: Perceptual Evidence

Audio 'ba' + Video 'ga' lips = Perceived 'da'

The brain automatically fuses audio and visual speech signals. Even knowing the trick, you cannot suppress the illusion. This proves the human brain has a built-in audio-visual fusion mechanism — AI systems that ignore this miss how the world actually works.

## Scale of Free Supervision

YouTube content uploaded: 500+ hours per minute

Video clips available: Billions (essentially unlimited)

Labelling cost: \$0 — inherently paired by physics

Signal type: Causal: visual event → audio event

Model quality vs scale: Scales with video data, not annotations

**Key insight:** The most abundant free supervisory signal ever discovered exists in ordinary internet video. This makes audio-visual learning fundamentally different from supervised learning — you need data, not annotations.

## SECTION 2 — SELF-SUPERVISED LEARNING

# Self-Supervised Audio-Visual Pre-Training

*Learning without labels: use the synchrony of audio and video as the supervisory signal*

# Three Pretext Tasks for Audio-Visual SSL

*Self-supervised learning defines tasks whose labels come from the data itself — no human annotation*

1

**Audio-Visual Correspondence (AVC)** Binary classification: does this audio come from the same video clip as this video frame? Positive pairs from the same clip; negative pairs from different clips (automatically generated). The model learns: what sounds go with what visuals.

2

**Temporal Audio-Visual Synchrony (AVS)** Regression task: predict the temporal offset (in seconds) between audio and video streams. Forces the model to learn fine-grained temporal alignment. Harder than AVC — requires understanding the dynamics of both streams.

3

**Masked Audio-Visual Modelling (MAVM)** Mask patches in the visual stream and/or spectral bands in audio. Predict the masked tokens using information from the other modality. This is AVHuBERT's approach — forces cross-modal reconstruction and is extremely data-efficient.

**Why these work:** All three pretext tasks are constructed automatically from unlabelled video. The model must develop deep audio-visual representations to solve them — these representations transfer to downstream tasks (ASR, VQA, retrieval) with far less labelled data.

# InfoNCE: The Contrastive Loss for Audio-Visual Learning

*Push matching audio-video pairs together; push non-matching pairs apart — in a shared embedding space*

$$\mathcal{L} = -(1/N) \sum_i \log \left[ \frac{\exp(\text{sim}(v_i, a_i)/\tau)}{\sum_j \exp(\text{sim}(v_i, a_j)/\tau)} \right]$$

*$v$  = visual embedding |  $a$  = audio embedding |  $\text{sim}$  = cosine similarity |  $\tau$  = temperature hyperparameter*

**Positive pair ( $v_i, a_i$ ):**

Same video clip — audio and visual streams are temporally aligned. The loss pushes their embeddings close together (high cosine similarity).

**Negative pairs ( $v_i, a_j$ ),  $j \neq i$ :**

Different video clips — audio from one clip paired with visual from another. Every other example in the batch is automatically a negative.

**Temperature  $\tau$ :**

Controls the sharpness of the distribution. Low  $\tau \rightarrow$  sharper focus on hard negatives, but training unstable. Typical values: 0.07–0.1. Learned parameter in CLIP.

**Batch size effect:**

Larger batch = more negatives per anchor = harder negatives = stronger learning signal. CLIP uses batch size 32,768 across multiple GPUs for this reason.

**NTXent is symmetric:** Compute loss in both directions: audio-to-visual and visual-to-audio. Sum them. This ensures the shared embedding space is symmetric — you can retrieve audio from visual queries AND visual from audio queries equally well.

## SECTION 3 — KEY MODELS

# AVHuBERT and AudioCLIP

*The two most important audio-visual pre-training models and their complementary approaches*

# AV-HuBERT: Masked Audio-Visual Pre-Training

*Shi et al. (Meta AI, 2022) — applying the masked prediction paradigm to audio-visual speech recognition*

Architecture: ResNet visual frontend (lip video frames) + convolutional audio frontend (spectrogram) → feature fusion → Transformer. Randomly masks input features (lip patches or audio spectral bands). Predicts discrete cluster labels (k-means pseudo-labels from offline clustering) for masked positions.

## Training Data

1,000 hours of unlabelled video: LRS3 (BBC lip reading) + VoxCeleb2. No transcription labels needed for pre-training — only speaker identity (for contrastive loss variants).

## Pre-training Objective

Predict the k-means cluster assignment of masked frames using context from both audio and visual streams. Forces the model to perform cross-modal reconstruction — if audio is masked, use video; if video is masked, use audio.

## Key Results

State-of-the-art visual speech recognition (lip reading) using only 30 hours of labelled data. Audio-visual ASR degrades gracefully with noise: -10 dB SNR reduces WER by only 15% vs 250% for audio-only model.

## Engineering Insight

Masked cross-modal prediction is extremely data-efficient. 1,000 hours unlabelled + 30 hours labelled outperforms fully supervised models trained on hundreds of labelled hours. The masking forces the model to actually use both modalities.

**Application:** AV-HuBERT enables lip reading and noise-robust ASR for accessibility — real-time captioning for deaf users in noisy environments, video conferencing enhancement, silent speech interfaces.

# AudioCLIP: Three-Way Contrastive Learning

*Guzhov et al. (ESC 2021) — extending CLIP's success to a simultaneous audio-image-text contrastive objective*

Architecture: Three encoders — CLIP ViT image encoder (pretrained, partially frozen), CLIP text encoder (pretrained), CNN14 audio encoder (trained from scratch). Three pairwise contrastive losses: audio-image, audio-text, image-text. Final loss = average of all three. Trained on AudioSet (2M clips, 527 sound categories).

## Zero-Shot Audio Classification

Classify audio events from text descriptions without seeing audio training examples. Text prompt: 'sound of a car engine starting'. AudioCLIP retrieves matching audio clips. No audio class labels required.

## Audio-to-Image Retrieval

Find images matching an audio clip. Input: 5-second recording of rain. Output: ranked images of rain, umbrellas, wet roads. Enables audio-based visual search — 'find me images that sound like this'.

## Text-to-Audio Retrieval

Find audio clips matching a text query. Input: 'sound of thunder'. Output: ranked audio clips by cosine similarity in shared embedding space. Powers audio asset discovery for content creators.

## Cross-Modal Embedding Space

Audio, image, and text all occupy the same 512-dimensional space. You can compute audio-text similarity directly without needing image as an intermediary. Enables emergent cross-modal reasoning not explicitly trained for.

**Foundation:** AudioCLIP is the foundation for audio-visual search engines, content moderation systems, and music information retrieval. Later extended by ImageBind (Meta, 2023) which aligned 6 modalities via images as the binding modality.

## SECTION 4 — TEMPORAL ALIGNMENT

# The Core Engineering Challenge

*Temporal misalignment corrupts audio-visual learning — SyncNet, filtering, and soft alignment strategies*

# Temporal Alignment: Problems and Solutions

*Video encoding delays, camera cuts, and background audio corrupt the natural audio-visual pairing*

**Video encoding delay:** Audio and video may be off by 100–500 ms due to codec processing. At 16 kHz audio and 25 fps video, even a 1-frame video delay means the audio is already 40 ms ahead of the frame it corresponds to.

→ Sample audio within  $\pm 0.5$  sec of the video frame timestamp. Train the model to be robust to small temporal offsets (within  $\pm 100$  ms) via data augmentation.

**Camera cuts:** Video editing cuts while audio continues. A cut in the middle of a sentence pairs audio of word A with video of a completely different scene. This is a false positive pair.

→ Scene cut detection (PySceneDetect) before creating training pairs. Discard pairs that span a detected scene cut.

**Background noise / off-screen sounds:** Audio contains sounds from off-screen sources (music, ambient sound) that have no visual correspondence in the frame. Model learns spurious correlations.

→ SyncNet: compute lip-audio sync score using a separate pre-trained lip-sync model. Filter training pairs below sync threshold of 5.0.

**Edited / narrated content:** Documentary narration, dubbed video, screen recordings — audio and video are deliberately de-coupled. Standard audio-visual correspondence assumption fails entirely.

→ Train/domain classifiers to detect narrated content. Filter by speaker face visibility (dlib face detector + lip motion threshold).

**Rule of thumb:** Invest 40% of your audio-visual project effort in data filtering. High-quality alignment filters (SyncNet + scene cut detection + face visibility) improve model quality more than architecture changes.

# Audio-Visual Models: A Comparison

*The evolution from simple correspondence models to universal six-modality embedding systems*

| Model              | Year | Pre-training        | Modalities       | Key Application                 |
|--------------------|------|---------------------|------------------|---------------------------------|
| AVC (Arandjelovic) | 2017 | Correspondence      | Audio + Video    | Sound source localisation       |
| SyncNet            | 2016 | Lip-sync detection  | Audio + Lips     | AV sync quality filtering       |
| AudioCLIP          | 2021 | Contrastive (3-way) | Audio+Image+Text | Zero-shot audio classification  |
| AV-HuBERT          | 2022 | Masked prediction   | Audio + Lips     | Visual speech recognition       |
| ImageBind (Meta)   | 2023 | Contrastive         | 6 modalities     | Universal cross-modal retrieval |
| VideoPoet (Google) | 2024 | Token prediction    | Audio+Video+Text | Audio-visual generation         |

**Trend:** Each year, more modalities are unified into a single embedding space. ImageBind (2023) aligns 6 modalities via images as the binding modality — enabling retrieval between any pair of modalities without direct paired training data between them.

# Key Takeaways

*The six essential points from this concept*

**Free supervision from video synchrony** Physical causality (visual event → audio event) provides virtually unlimited training signal. Every video is a labelled pair. No annotation cost.

**Three SSL pretext tasks** Audio-Visual Correspondence (same/different clip), Temporal Synchrony (offset regression), Masked Audio-Visual Modelling (cross-modal reconstruction). All labels derived automatically.

**InfoNCE is the dominant loss** Push same-clip (positive) pairs together, push different-clip (negative) pairs apart. Larger batch = harder negatives = stronger signal. Temperature  $\tau$  controls sharpness.

**AVHuBERT: masked cross-modal prediction** 30 labelled hours + 1,000 unlabelled hours of video → SOTA lip reading. Masking forces genuine cross-modal reasoning.

**AudioCLIP: three-way contrastive** Audio + Image + Text in one shared space. Enables zero-shot audio classification, audio-to-image retrieval, and text-to-audio search from a single model.

**Temporal alignment is the engineering bottleneck** SyncNet filtering + scene cut detection + face visibility checks determine data quality more than architecture. 40% of effort should go to data filtering.

**Note:** Next: AI-MM-IN-TH-000012 — Training Multimodal Systems (3-stage pipeline, objectives, LLaVA-1.5 recipe)

AI-MM-IN-TH-000011 Complete

# Audio-Visual Learning

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